Fluids and Electrolytes: CA-1

Part 1: The slides that are now a handout

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Debates

1. **Liberal vs. Restrictive Fluid Strategies:**
2. **The Role of Goal-Directed Fluid Therapy:**
3. **Crystalloids vs. Colloids (Specifically Albumin) in Surgical Patients:**
4. **Normal Saline vs. Balanced Crystalloids (e.g., LR, Plasma-Lyte):**
5. **The Utility of the 4/2/1 Rule for Fluid Deficit Calculation:**
6. **Preoperative Oral Clear Liquids Up to 2 Hours Before Surgery:**
7. **The Clinical Relevance of Dynamic Assessments of Fluid Responsiveness:**
8. **Routine use of a fluid "mini-bolus" to assess fluid responsiveness:**
9. **Early initiation of Massive Transfusion Protocols in Trauma Patients:**
10. **The Role of Intraoperative Blood Pressure as a Primary Goal for Fluid Management:**
11. **The Use of Esophageal Doppler for Intraoperative Fluid Guidance**
12. **The Value of Central Venous Pressure (CVP) Monitoring as a Guide to Fluid Resuscitation:**
13. **The Clinical Significance of Glycocalyx Damage in Fluid Management:**
14. **The Use of D5W as an Intravenous Fluid in Modern Anesthesia Practice:**
15. **Should Fixed Maintenance Rates for Fluid Administration Be Completely Abandoned?**
16. **The Use of Vasopressors as a Primary Treatment for Hypotension (Instead of Fluids) in Certain Surgical Populations:**
17. **The Evidence for ERAS Protocols in Significantly Improving Post-Operative Outcomes Based Primarily on Fluid Strategies:**

I generated this list of debate topics last year. I had thought about assigning teams to come up with a short Pro/con for each of these, but I was told that I would be SAFE reported. however, I think that these are some good questions to think about how you would answer if they came up on a board stem for you.

Electrolytes

Volumes of distribution:

- Potassium
- Sodium
- Calcium
- Magnesium
- Glucose
- Water

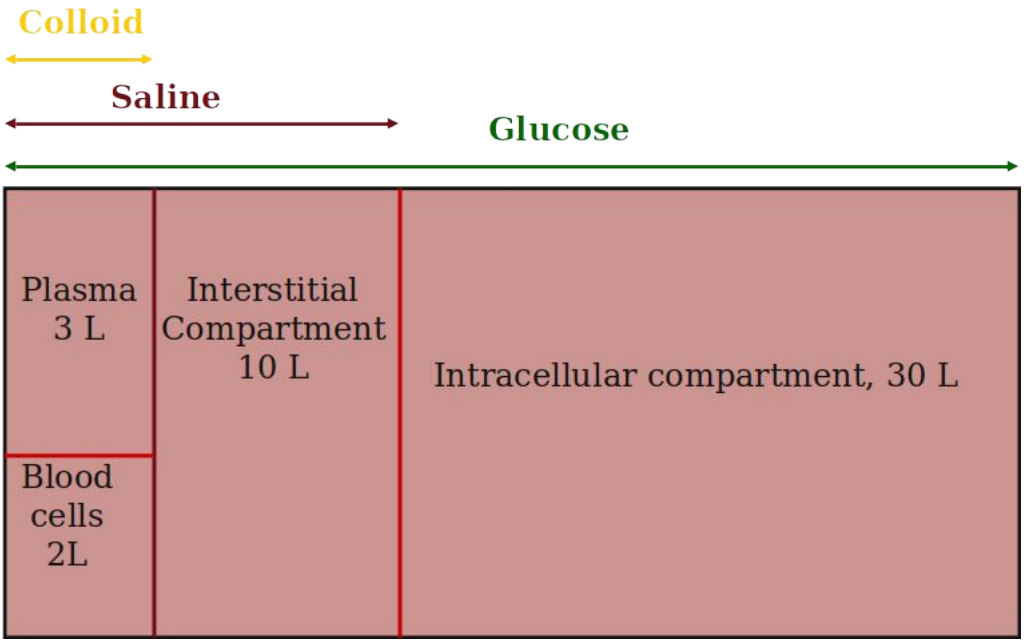
Deficiency / Supplementation

- Potassium
- Calcium
- Magnesium

I am much more interested in fluid, so Let's talk about electrolytes first. I am not going to read a table at you, but you should be familiar with the volume and distribution of each of these.

you should also know the cardiovascular effects of deficiencies and high levels, as well as the typical Doses and effects of usual strategies .

You are reasonably likely to be quizzed and tested on the electrolyte content of each fluid



As a reminder, this is what the content outline means for volumes of distribution. even though this is well understood to not actually be true, it will be true for your test.

Fluids compositions

	Normal Saline	Lactated Ringer	5% dextrose	Plasma-Lyte
Na	154	130	0	140
Cl	154	109	0	98
K+	0	4	0	5
Ca++	0	3	0	
lactate	0	28	0	
glucose	0	0	50 g/L	(21 kcal/L)
osmolality	308	275	252	294

Boom, there's a table that I'm not going to read at you. you will just have to memorize this thing on your own time. the memorable things are the Supra physiologic chloride concentration in normal saline, the small amount of potassium and calcium in lactated ringers, any absence of potassium in plasmalyte.

for your exams, you should continue to believe that any solutions such as lactating ringers which contains Calcium is incompatible with blood products, even though that is not really true for practical purposes.

The key questions with fluid

- How much, if any, IV fluid does this person need. Harms?
  - In the OR vs ICU
  - Edema issue: burn, plastics, neuro, HPB
  - Major risk of pulm complications with overload: thoracic
  - Increases portal pressure or CVP: GI, liver txp
  - Major trauma: damage control, want mildly hypotensive
  - Peds: many separate considerations
- What fluid should I give them
  - Crystalloid, colloid, blood
- How fast should I give it?

Fluid benefits / harm

Increased cardiac output / decreased vasoconstriction / sympathetic outflow

- Strong evidence that aggressive practice shifts to favor vasopressors have increased AKI

- Decreased PONV reasonable quality evidence, fairly strong effect at large volumes (20 ml/kg RR ~ 0.6)

Slower return to bowel function

Tissue edema -> poor wound healing , anastomotic leaks

Hemodilution

Coagulopathy

Lung water

So why are we talking about this. obviously there are many patients who need fluid, for example because their blood has been falling out. for the large group of patients who has not had significant blood loss during their surgery though there are still substantial changes in cardiac preload which could be treated with fluid administration. the body is reasonably good at sensing when it is Volume depleted, and uses changes in sympathetic tone as well as renal clearance to try to make up for that. those endogenous mechanisms aren't without cost though, and there may be a benefit to reducing acute kidney injury and improvements to other post-operative outcomes by Giving fluids.

however, there are also very well demonstrated harms to fluid administration. one of the strongest and most replicable associations To mortality in the critical care literature is net fluid balance. in the purely operative phase, it's well appreciated that excessive fluid administration and resulting tissue edema have adverse effects on wound healing. fluid overload above the ability to compensate with increased stroke volume also leads to pulmonary edema. in settings in which very large volumes of crystalloid are administered, such as old style trauma resuscitation, there are also concerns about diluting the major components of blood.

Fluid deficit

Calculating deficit - not recommended, but might show up on test:

- 4/2/1 rule: 4 cc/kg/hr for the first 10 kg, 2 cc/kg/hr for the second 10 kg, and 1 cc/kg/hr for every kg above 20. ~1.2 cc/kg/hr for adults (PMID 30238969 vs 17856681)

- IBW? Lean? Good luck.

- Minimal *intravascular* change for 10 hrs.

Minimizing deficit: minimize fasting

- No rationale for > 2 hrs clears without known gastroparesis (glucose slower)

- Gastric US to verify

- 500 mL sugary drink seen in ERAS has minimal evidence of benefit

Other sources of intravascular volume loss

- Bowel prep, bleeding, exposure, edema, burns

So we want to administer the right amount of fluids to each patient. one piece of that is figuring out if someone is already depleted and then replacing that. because of overnight fasting for first start cases, historically there was an emphasis on loading patients to make up for Losses incurred during fasting. Calculating a fluid deficit is a largely obsolete Concept in most surgery. however, it may still show up on your test and so you should know these basic formulas. interestingly recently somebody did replicate the classic studies from the 1800s and this second paper is too patients in the 1930s establishing that adults do lose about one mil per kilogram per hour in an overnight fast. that data is all in healthy lean adults, and there is very little to guide what changes in the typical overweight population.

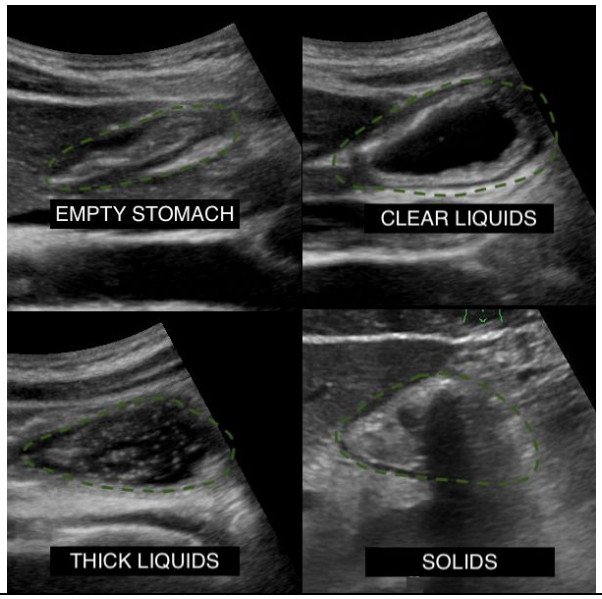
more recently, there has been a larger emphasis on just minimizing the duration of going without liquids. water on average is 50% cleared from the stomach in about 12 minutes for patients without gastroparesis, glucose containing Solutions are a little bit slower than that. still, there is minimal reason to not allow clear liquids before surgery, and universally we now encourage liquid intake up to 2 hours prior. for patients with gastroparesis, you should be comfortable doing a gastric ultrasound to verify that the stomach is relatively empty. One Hot Topic in the community right now is the effect of glp-1 receptor antagonists on gastric emptying, with reports of grossly full stomachs despite appropriate fasting but no obvious impact on aspiration related complications.

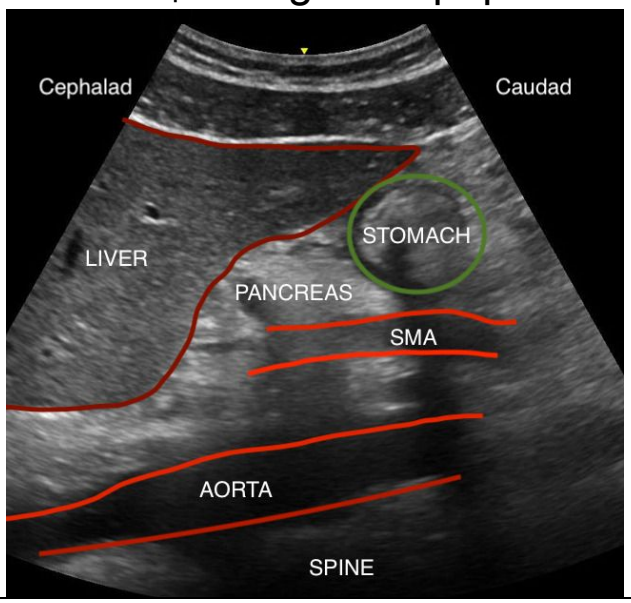
There are certainly other populations where you might suspect that the patient is going in volume depleted. such as those who have been bleeding, those who have a physiologic mechanism for edema, Burns, those who have been exposed for quite

some time. there's also some effect of bowel prep solutions, although it is less than it used to be

Gastric US

http://www.usra.ca/regional-anesthesia/specific-blocks/pocus/gastric.php





https://www.ncbi.nlm.nih.gov/books/NBK580524/

Fluid deficit

Fixed Maintenance losses: also fairly obsolete concept:

- Minimal tissue trauma (ex. herniorrhaphy): 2-4 cc/kg/hr
- Moderate tissue trauma (ex. cholecystectomy): 4-6 cc/kg/hr
- Severe tissue trauma (ex. bowel resection): 6-8 cc/kg/hr

Minimizing losses

- HME
- lap / robot
- fluid in field

You also see questions about maintenance losses. this again is not a very relevant concept anymore, but it is relevant to think about expectations for especially long surgeries with lots of exposure and evaporative fluid loss, such as a large spine case or large open abdominal case. Instead, we are much more likely to think about dynamically assessed losses and assessing patient responsiveness, which we will get to in a little bit. The above number are often cited, but are likely overestimates in an era in which we are not blowing dry air into the patient (HME), are keeping operative surfaces pretty moist, and are using much smaller exposures. Notably minimally invasive surgeries blow dry air into the abdomen, but net circulation is modest and there is much less tissue disruption leading to lower fluid loss.

Real Losses

- Bleeding
- UOP
- Evaporative
- Tissue edema
- open abdomen -> loss of abdominal pressure -> fluid translocation

And this is not to say that these real sources of loss of intravascular volume don't occur, just that fixed maintenance ratios are not a very good strategy.

Temporary Decreased Filling

- Almost all anesthetic drugs cause vasodilation, sympathetic blockade esp spinal. *Venous* as well as arterial
  - EEG guided anesthesia reduces vasopressor / fluid requirements
- Preloading with IVF for neuraxial now uncommon. Ephedrine nice choice (maintain HR).
- Abdominal insufflation, vein compression
- Positive pressure ventilation

They're also lots of reasons for a patient to have a temporary decrease in preload. some of these are going to persist throughout the surgery, but then the eliminated when the patient wakes up, others are quite transient during the surgery. In cases like these, it's important to ask whether or not the temporary benefit of overcoming a temporary loss of cardiac filling is worth the price paid over the next several days for volume overload. of course, to maintain perfusion the other option is typically vasopressors or less commonly inotropes, which have their own risks

Fluid strategies

"Liberal" - load 1-2 L and 6 mL/kg/hr + replace blood loss

"Restrictive" - minimal load + replace blood loss

Goal directed

ERAS

- Minimize deficit, aggressive preop PO fluids
- Euvolemia
- Targeted fluid administration in major surgery: NIPPV, doppler, quantitative blood loss
- Shorter surgeries with anticipated early PO: err on side of hypovolemia

There are a couple of strategies that you will classically see and hear discussed. however like most things in the anesthesia and critical care literature they are extremely poorly defined and not operationalized in even vaguely the same way across studies.
the more traditional approach was what is now called a liberal fluid strategy. that used a deficit calculation like we saw earlier to come up with the idea that everybody needs one to two liters of crystalloid, and then a pretty stiff maintenance rate to make up for surgical losses, along with replacement of blood losses with crystalloid or colloid as we will discuss later. this tended to lead to pretty enormous volumes of crystalloid when unthinkingly carried out, and as a reaction to the significant edema that these patients had in the '90s and 2000s (the perioperative michelin man) there were many studies of a restrictive fluid strategy. those typically have minimal or no load at the start of the case, low (such as one ml per kilo per hour) maintenance and replacement of blood losses as well as bonuses necessary for hemodynamic changes. If it shows up on your exams, you should assume that a liberal strategy is bad and that you should not do it. however, as we will see the literature is not so clear-cut. it mostly shows that there is not much difference between the two.

Again in the '90s and 2000s to 2010s there were many studies on what is usually labeled goal-directed therapy. in goal-directed therapy some target metric is chosen, and fluid administered essentially only as long as it is improving that metric.
goal-directed therapy was frequently included in early recovery after surgery bundles along with a generally restrictive fluid goal and aggressive attempts to increase oral intake. While eras is very widespread and very popular, because they are bundles of

interventions, typically the studies do not really tell us what if any of the elements are working even if they are otherwise high quality studies.

Determining Fluid Status

Labs usually a late indicator

- hgb mostly for overload
- Lactate / SVO2 for general hypo-perfusion

Clinical signs:

- BP and HR easily confounded, not reliable in many patients
- UOP < **0.3 or 0.5** mL/kg/hr if **sustained**
- capillary refill (crit care: 41159835, 30772908)

CVP largely useless: supply vs demand

What are some of the older ways of determining that a patient needed fluid? first, one could look at lab values. Hemoglobin certainly can change with bleeding, although it is important to remember that it takes several hours for hemoglobin levels to re-equilibrate after acute blood loss. dilutional anemia is certainly something that one can encounter, and is important to remember that not all anemia needs blood. Hemoconcentration in the presence of vasopressors should usually be worrying. in the OR and in the emergency department you will often see lactate and central venous oxygen saturation used as markers of need for fluid, although they are really markers of hypoperfusion. if you have already decided that the patient does not have a contractility problem, ischemic tissue, or other special reasons to have an elevated lactate, it is a reasonable sign that you need to do something to increase cardiac output. however, central venous oxygen saturations can be paradoxically high in vasoplegic States in which there is normal or higher than normal cardiac output but it is not being adequately distributed, such as sepsis or end stage liver disease. I'm not saying that those patients necessarily need fluid but that evaluating the results is a little more complicated.

available to us in every OR are vital signs. and while they certainly are useful and you all already know how to evaluate them, they are unreliable in many settings. for example, blood pressure and heart rate also depends on nociceptive response, Adrenergic drugs, and parasympathetic outflow from stimuli such as insufflation. There are some very special situations in which central venous pressure is useful, but usually in the other direction. that is, once you see elevations in central venous pressure, that means that the patient's cardiac output is not augmenting to sufficiently

absorb the increase in preload. in unusual circumstances, patients with elevated central venous pressure may still need volume. however, the other implication that low central venous pressure implies either fluid responsiveness or a need for volume is definitely not true. for example, because I am generating negative intrathoracic pressure with every inspiration, my CVP is probably a negative number right now, but I do not need a fluid bolus. there are similar arguments to be made for pulmonary capillary wedge pressure, but because you are functionally only going to see pulmonary artery catheters in cardiac surgery, I will defer that discussion to them.

› Perioper Med (Lond). 2015 Apr 10;4:3. doi: 10.1186/s13741-015-0014-z. eCollection 2015.

Perioperative fluid therapy: a statement from the international Fluid Optimization Group

Lais Helena Camacho Navarro [1], Joshua A Bloomstone [2], Jose Otavio Costa Auler Jr [3], Maxime Cannesson [4], Giorgio Della Rocca [5], Tong J Gan [6], Michael Kinsky [7], Sheldon Magder [8], Timothy E Miller [6], Monty Mythen [9], Azriel Perel [10], Daniel A Reuter [11], Michael R Pinsky [12], George C Kramer [7]

Affiliations + expand

PMID: 25897397 PMCID: PMC4403901 DOI: 10.1186/s13741-015-0014-z

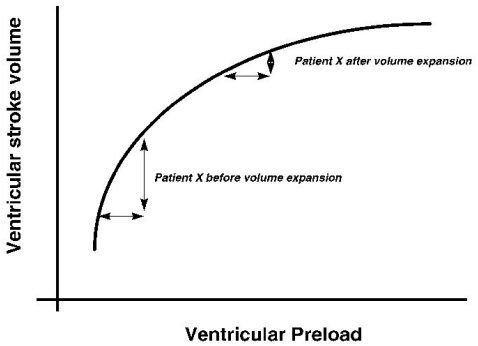
Free PMC article

As we go into specific modern methods, this paper is getting a little old but still pretty well represents what I would summarize as my beliefs on the efficacy of different techniques. as you read it there is a lot of humility, since they acknowledge that many of the things we would like to be true don't actually work in practice

Dynamic fluid status: Respiratory-variation-type

Starling-curve based approach

- Arterial PPV, esophageal doppler SVV, pleth V
- invalid in AF, open chest, low TV, high PEEP, less useful in high abdominal pressure, RV problems, marked tachycardia
- Hard to calibrate, grey zone



So what about Dynamic assessment of fluid status. there are a couple of Technologies that purportedly allow the anesthesiologist to determine if the patient is likely to increase their cardiac output in response to fluids, which is how we think of fluids as helping the patient.

The most widely used set of approaches comes from respiratory variation and the Starling curve. In case you have somehow forgotten, the Frank Starling law relates some measure of ventricular preload on the x-axis to stroke volume on the y-axis, and is non-linear with a steep increase at low preloads and a small slope at higher preloads. if you want to start a fight in the ICU, you can ask whether or not the slope is ever negative. all the respiratory variation methods work by assuming that positive inspiratory pressure decreases venous return temporarily, and the release of positive inspiratory pressure increases venous return temporarily. therefore, when averaged over a few respiratory Cycles for patients who are at the low end of ventricular preload (and therefore able to increase their stroke volume with fluid administration) there will be high variation in something that correlates to stroke volume. On the other hand patients who are in the relatively flat/insensitive part of the Starling curve, will have low variability measures of stroke volume* over a respiratory cycle.

that "something correlated to stroke volume " is usually arterial line pulse pressure, giving the classic pulse pressure variation measure. however you can apply the same theory to esophageal Doppler based estimates of stroke volume, signal variation on a plethysmograph, a finger-cuff based estimate of stroke volume, thoracic impedance, pulse-waveform based stroke volume, or many other similar tools.

While this is a really useful tool, it comes with a lot of limitations. first you need an absence of things like atrial fibrillation which cause very irregular stroke volumes. you need the change in venous return to be pretty large, meaning that it doesn't work with the low tidal volumes that we typically use now, open chests, very high peak airway pressures. in laparoscopic or robotic cases with high abdominal insulation, the change is also often too low. it can be difficult to get this technique to work if the right ventricle is not functioning well, or the heart rate is very High, limiting the opportunity for venous return (filling time limited).

Although I mentioned many stroke volume correlates, arterial line pulse pressure is the most widely used, because it is easy to understand and troubleshoot. if you have substantial noise or inability to accurately measure any of the other tools, then obviously they will not work well. for example, esophageal Doppler is a very useful technique in patients that don't require a nasogastric tube, but it takes some skill to position correctly. the pleth-based tools are less reliable because of differences in finger thickness and nail character, and the impedance measures are just unreliable to start with. these techniques all come with a pretty substantial Gray Zone in which it is not highly sensitive or specific, although very high values tend to be specific. They're also some confusing clinical situations, such as high post pressure variation in the absence of hypotension which suggests that although the patient might increase their cardiac output, they have successfully compensated to maintain their blood pressure. whether or not it benefits them to increase their cardiac output with normal blood pressure is harder to say.

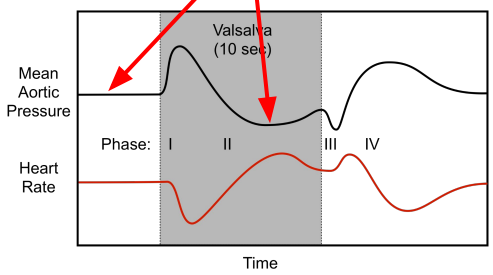
Non-PPV-like

Delta CO-like with PLR or trendelenburg

Delta with valsalva

Delta with end-expiratory pause

Compare these



PLR / EEOT / fluid based increase in preload with a small change in CO

Patient is near inflection point

Valsalva based decrease in preload with a large change in CO

A very similar starling curve argument that does not require beat to beat monitoring or consistency is using a valsalva to decrease venous return or a PLR (ICU) or steepening trendelenburg positioning to increase it. this is a lot less standardized in the literature, but I will not uncommonly use it in patients with atrial fibrillation to verify a hypothesis that they are volume responsive. note that valsalva is the opposite of a passive leg raise, and not the right direction. A 30% decrease in systolic pressure with valsalva indicates fluid responsive. If pulse pressure variation is available, 50% increase in PPV is a very good predictor, better than baseline PPV. A passive leg raise temporarily increases venous return and You observe a positive response. a valsalva temporarily decreases venous return, and you observing negative response. just because decreasing venous return decreased cardiac output, does not imply that you will get much of an increase with additional volume, although that tends to be true.

An end-expiratory occlusion test (EEOT) is the opposite of the valsalva and like looking at “phase 4” in the above diagram. Pause the vent at low peep after complete exhalation (some ventilators have an expiratory pause button). Having unloaded the positive thoracic pressure impeding venous return, in fluid responsive patients you should see a CO increase of 5% or more. EEOT requires patient not to initiate!

Dynamic arterial elastance

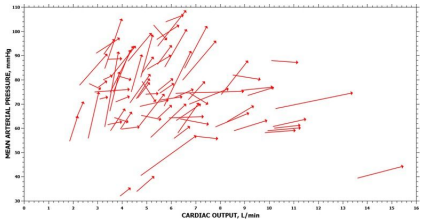
High SVV predicts *CO* increase with fluid better than MAP

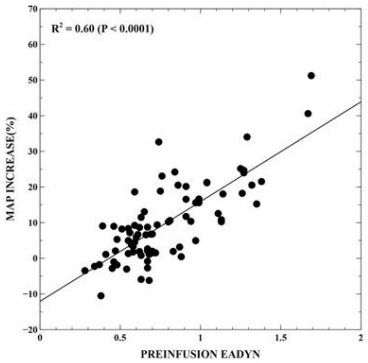
The relationship between PPV and SVV is itself not uniform

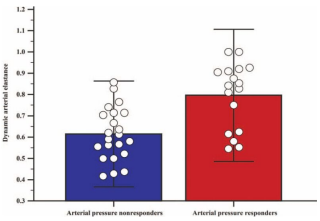
High EaDyn (= PPV/SVV) is a good predictor of response to both IV fluids and norepinephrine in *critically ill adults.*

The literature is pretty small, and less convincing intraop

Physiology a little wooly, probably intact vascular tone in higher EaDyn

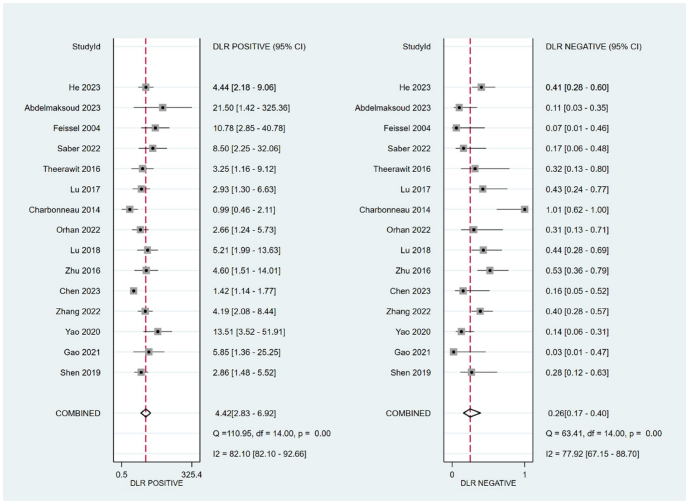


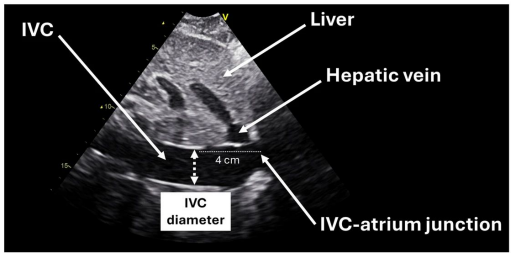


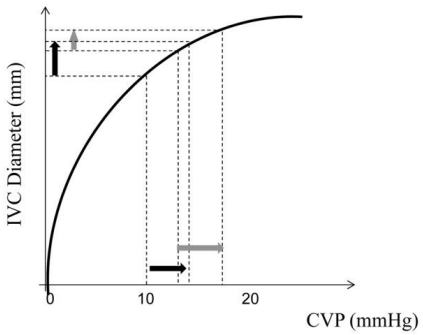


https://pmc.ncbi.nlm.nih.gov/articles/PMC4271484/

Non-respiratory variation: IVC diameter / distensibility







IVC diameter correlates to CVP, which is not very useful except in narrow populations. Measurement seems easy, but can be hard to get m-mode lined up or a perpendicular slice.

Respiratory variation does not require beat-to-beat hemodynamics, has a similar shape to starling-law PPV argument

IVC collapsibility in negative pressure vent is normal, lack suggests fluid overload.

IVC distensibility [(dmin-dmax)/dmin] in positive pressure is complicated: ability of fluid in deep veins to quickly flow (reserve), IVC capacitance, abdominal pressure, ability of right heart to quickly fill, how much vent pressure obstructs flow. 15%-18% cutoffs frequently used, works ok, usually labeled as “moderately” predictive because of the variability among populations. IVC distensibility also probably does not work in an open chest.

Venous excess ultrasound score (VExUS) is a related multi-target assessment of venous *congestion* to stop additional fluids.

Just give some and see what happens

Mini-bolus fairly effective

Needs to be rapid (< 5 min)

ΔSV (4-6%)

ΔSVV (3%) more accurate

Observational Study > Br J Anaesth. 2015 Sep;115(3):449-56. doi: 10.1093/bja/aev222. Epub 2015 Jul 6.

Decrease in pulse pressure and stroke volume variations after mini-fluid challenge accurately predicts fluid responsiveness†

J Mallat [1], M Meddour [2], E Durville [2], M Lemyze [2], F Pepy [2], J Temime [2], N Vangrunderbeeck [2], L Tronchon [2], D Thevenin [2], B Tavernier [3]

Affiliations + expand

PMID: 26152341 DOI: 10.1093/bja/aev222

Smaller (100mL) avoids HR effects, 250 mL has a higher response rate. 500 mL often used in crit care.

I think this works, but there is weird mathematical coupling in some studies, so hard to say how well.

So the best way to try and figure out if someone is fluid responsive Is to do so in retrospect. I have many times asked the clinician in the room to try giving some volume, and if it worked keep doing it until it stopped working. the problem with that sort of simplistic motto is that there may be other things happening at the same time, often leading people to falsely believe that fluid didn't help, not realizing that because the patient was for example still bleeding they were just treading water. you will see strategies of the fluid challenge or mini fluid challenge (normally meaning 100 mils to 200 mL as a test) looking for changes in crudely blood pressure (or vasopressor requirement) but with much better accuracy if you can use something like one of the tools mentioned above that correlates reasonably well to cardiac output. even better, if you are in a patient state where pulse pressure variation is reasonably accurate, changes in pulse pressure variation or one of the analogs is even more accurate than using the current pulse pressure variation.

Complementary: track absolute CO

Pulse contour etc.

Finger clamp.

Impedance

CO_2 elimination

Aortic VTI (esophageal doppler)

TTE/TEE LV SV / LVOT VTI

Obviously need for IV fluids isn't the only or even main reason to have decreased CO, but monitor that is more like CO lets you track the response to respiratory variation, a mini-fluid challenge or other moves without worrying about reflex changes in arterial tone mitigating changes in BP, which is common.

Also on the market and used clinically are devices like pulse contour analysis or pulse waveform analysis and finger clamp based methods for non-invasive cardiac output measurement. although they tend not to be super well calibrated they are reasonably effective for trending, and are used in some goal-directed therapy protocols as the target. Carbon dioxide elimination is another promising tool for estimating short-run changes in CO.

True lithium dilution lidco cardiac output is available but much less commonly used than the pulse wave analysis from lidco. all the pulse contour and pulse wave methods are built with a lot of assumptions about coupling between peripheral arterial resistance and central resistance, and are as best I'm aware not really validated in states like sepsis, liver disease, post bypass vasoplegia, neurogenic shock Etc in which we would be really interested in having this data available.

esophageal Doppler is a much less assumption written tool, but as I said can be tricky to use in practice.

in the modern era, with sick high risk patients, if you have the skills (you should!), you can of course put in a transesophageal echo (or transthoracic), and directly estimate

lvedv or ejection fraction. obviously there are many limitations to transthoracic echo including whether or not you have access to the chest and the patient has good windows. Transthoracic Echo also needs more practice and probe manipulation skill than transesophageal (IMO), making it less desirable for a rescue application.

Risk vs Benefit

- Anastomosis (esophagectomy, pancreatectomy, small bowel)
- Delayed bowel function
- Increased blood loss (liver resection)
- tissue perfusion (flaps)

So far the goal directed therapy has been only about potential benefit in terms of raising cardiac output. however you still need to consider the risk benefit calculation even if there is some benefit. for example, you may very well tolerate a degree of suboptimal cardiac output if the resulting fluid administration would cause tissue edema that threatens an anastomosis, such as in pancreas surgery, increases blood loss such as in liver surgery, or threatens tissue perfusion such as in flap cases.

What fluid should I give?

Do they need blood? Other blood products?

- In particular cases, should blood be given earlier?
  - Exsanguinating trauma, avoid crystalloid > 1.25 L. PAMPer suggest benefit of pre-hospital plasma, RePHILL did not
- What ratio of components should be used?
  - 1:1:1 (1 bag plt = 6) a reasonable default past 0.5 BV, limited evidence vs 2 rbc
  - With slow but ongoing losses, reasonable to target components
- When should whole blood be preferred?
  - Reasonable evidence in military trauma, strong benefit in observational civilian trauma
- When should MTP be started?
  - Excellent evidence that MTP reduced mortality, costs
  - "massive blood transfusion for trauma" other similar scores. Activate early!

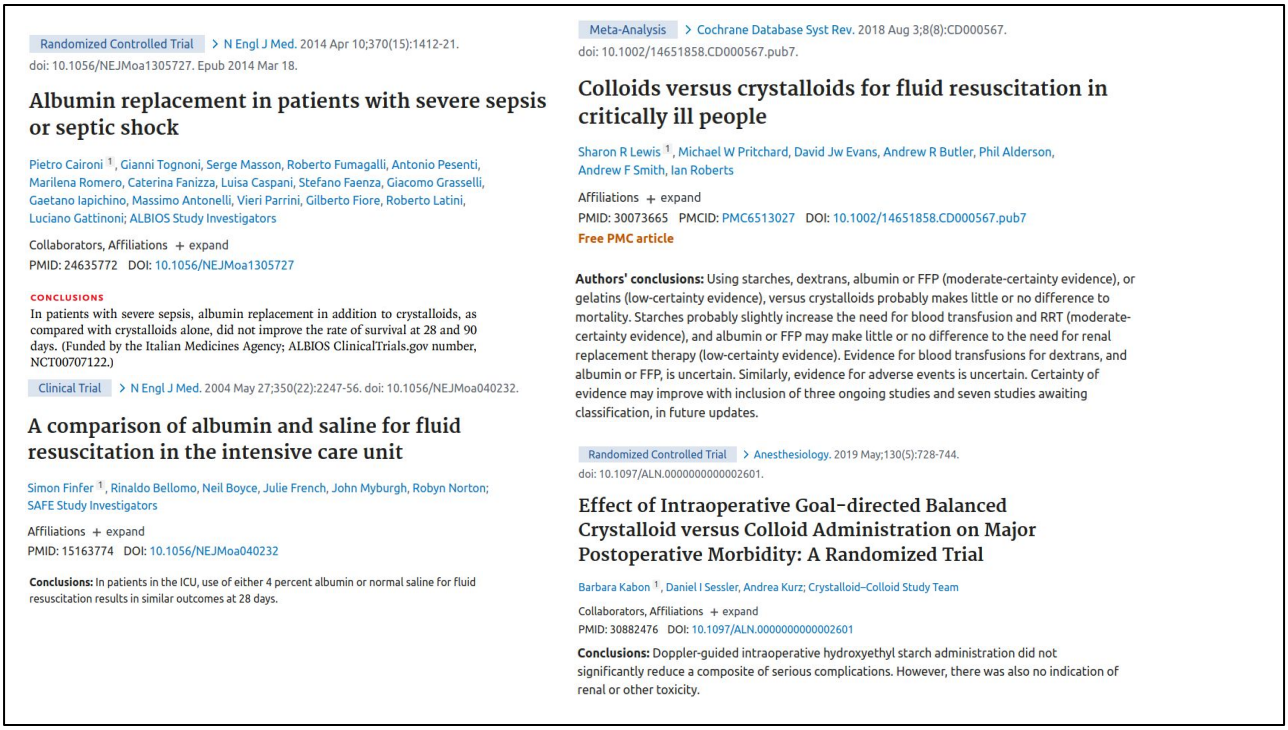
The second big question in fluid administration is you've decided that the patient needs some volume, what volume are you going to get them. the first thing to decide is do they need blood. Many of the questions around blood are separate talks, or at least I hope that they are. for example, you will generally see transfusion of red blood cells for hemoglobins of 7 to 8, with some variation around patients with significant cerebrovascular or coronary arterial disease. massive transfusion is also a separate topic, and there are good reasons for patients who are rapidly bleeding to skip other evaluative steps and jump straight to transfusion. There are also patients with dilutional anemia, who may or may not need any additional volume. other blood fractions fall under separate topics.

What fluid should I give?

How about some colloid?

The next question is if they don't need blood, do they need colloid or Crystalloid. Fortunately, other than dextrans administered purposely as an anticoagulant, you're only choice is human albumin in either 5% or 25% concentration. This is another good news bad news situation. the good news is that the boards answer is easy. outside of cirrhosis and large volume paracentesis, there is no group of humans who has been robustly shown to benefit clinically from albumin. depending on when the question writers were looking at the literature, there are defined subgroups of patients who were believed to be harmed by albumin. we will get into some more details on that in the next couple of slides, but it means that on a test you basically never have to worry about it be being the right answer. you may still get questions that are probing whether or not you know that the risk of acute injury from albumin is low.

however, when it comes to practical decisions in the operating room, you are going to encounter many different practice patterns, some of which use albumin heavily and some of which use it rarely if at all. again this is going to be a situation where you just have to read the room



Randomized Controlled Trial > N Engl J Med. 2014 Apr 10;370(15):1412-21.
doi: 10.1056/NEJMoa1305727. Epub 2014 Mar 18.

Albumin replacement in patients with severe sepsis or septic shock

Pietro Caironi [1], Gianni Tognoni, Serge Masson, Roberto Fumagalli, Antonio Pesenti, Marilena Romero, Caterina Fanizza, Luisa Caspani, Stefano Faenza, Giacomo Grasselli, Gaetano Iapichino, Massimo Antonelli, Vieri Parrini, Gilberto Fiore, Roberto Latini, Luciano Gattinoni; ALBIOS Study Investigators

Collaborators, Affiliations + expand
PMID: 24635772 DOI: 10.1056/NEJMoa1305727

CONCLUSIONS
In patients with severe sepsis, albumin replacement in addition to crystalloids, as compared with crystalloids alone, did not improve the rate of survival at 28 and 90 days. (Funded by the Italian Medicines Agency; ALBIOS ClinicalTrials.gov number, NCT00707122.)

Clinical Trial > N Engl J Med. 2004 May 27;350(22):2247-56. doi: 10.1056/NEJMoa040232.

A comparison of albumin and saline for fluid resuscitation in the intensive care unit

Simon Finfer [1], Rinaldo Bellomo, Neil Boyce, Julie French, John Myburgh, Robyn Norton; SAFE Study Investigators

Affiliations + expand
PMID: 15163774 DOI: 10.1056/NEJMoa040232

Conclusions: In patients in the ICU, use of either 4 percent albumin or normal saline for fluid resuscitation results in similar outcomes at 28 days.

Meta-Analysis > Cochrane Database Syst Rev. 2018 Aug 3;8(8):CD000567.
doi: 10.1002/14651858.CD000567.pub7.

Colloids versus crystalloids for fluid resuscitation in critically ill people

Sharon R Lewis [1], Michael W Pritchard, David Jw Evans, Andrew R Butler, Phil Alderson, Andrew F Smith, Ian Roberts

Affiliations + expand
PMID: 30073665 PMCID: PMC6513027 DOI: 10.1002/14651858.CD000567.pub7
Free PMC article

Authors' conclusions: Using starches, dextrans, albumin or FFP (moderate-certainty evidence), or gelatins (low-certainty evidence), versus crystalloids probably makes little or no difference to mortality. Starches probably slightly increase the need for blood transfusion and RRT (moderate-certainty evidence), and albumin or FFP may make little or no difference to the need for renal replacement therapy (low-certainty evidence). Evidence for blood transfusions for dextrans, and albumin or FFP, is uncertain. Similarly, evidence for adverse events is uncertain. Certainty of evidence may improve with inclusion of three ongoing studies and seven studies awaiting classification, in future updates.

Randomized Controlled Trial > Anesthesiology. 2019 May;130(5):728-744.
doi: 10.1097/ALN.0000000000002601.

Effect of Intraoperative Goal-directed Balanced Crystalloid versus Colloid Administration on Major Postoperative Morbidity: A Randomized Trial

Barbara Kabon [1], Daniel I Sessler, Andrea Kurz; Crystalloid–Colloid Study Team

Collaborators, Affiliations + expand
PMID: 30882476 DOI: 10.1097/ALN.0000000000002601

Conclusions: Doppler-guided intraoperative hydroxyethyl starch administration did not significantly reduce a composite of serious complications. However, there was also no indication of renal or other toxicity.

So in the critical illness literature the overwhelming balance of evidence does not favor colloid resuscitation. in the perioperative trials that we have, colloid is again not beneficial. there are some other landmark studies that if we were medicine residents we would go over, but the metanalysis on the top right there is reasonably up-to-date for this topic

Randomized Controlled Trial > Anesthesiology. 2016 May;124(5):1001-11.
doi: 10.1097/ALN.0000000000001051.

Effect of Exogenous Albumin on the Incidence of Postoperative Acute Kidney Injury in Patients Undergoing Off-pump Coronary Artery Bypass Surgery with a Preoperative Albumin Level of Less Than 4.0 g/dl

Eun-Ho Lee [1], Wook-Jong Kim, Ji-Yeon Kim, Ji-Hyun Chin, Dae-Kee Choi, Ji-Yeon Sim, Suk-Jung Choo, Cheol-Hyun Chung, Jae-Won Lee, In-Cheol Choi

Affiliations + expand
PMID: 26891150 DOI: 10.1097/ALN.0000000000001051

Conclusion: Administration of 20% exogenous albumin immediately before surgery increases urine output during surgery and reduces the risk of AKI after off-pump coronary artery bypass surgery in patients with a preoperative serum albumin level of less than 4.0 g/dl.

Randomized Controlled Trial > JAMA. 2022 Jul 19;328(3):251-258.
doi: 10.1001/jama.2022.10461.

Effect of 4% Albumin Solution vs Ringer Acetate on Major Adverse Events in Patients Undergoing Cardiac Surgery With Cardiopulmonary Bypass: A Randomized Clinical Trial

Eero Pesonen [1], Hanna Vlasov [1], Raili Suojaranta [1], Seppo Hiippala [1], Alexey Schramko [1], Erika Wilkman [1], Tiina Eränen [2], Kaapo Arvonen [1], Maxim Mazanikov [1], Ulla-Stina Salminen [3], Mihkel Meinberg [1], Tommi Vähäsilta [3], Liisa Petäjä [1], Peter Raivio [3], Tatu Juvonen [3], Ville Pettilä [1]

Affiliations + expand
PMID: 35852528 PMCID: PMC9297113 DOI: 10.1001/jama.2022.10461

Conclusions and relevance: Among patients undergoing cardiac surgery with cardiopulmonary bypass, treatment with 4% albumin solution for priming and perioperative intravenous volume replacement solution compared with Ringer acetate did not significantly reduce the risk of major adverse events over the following 90 days. These findings do not support the use of 4% albumin solution in this setting.

Cirrhosis: a bunch of studies

Flaps / anastamosis: ???

Ultrafiltration: a bunch of studies

Large volume resuscitation: ???

There have been some signals that for Some patient populations, but I am very skeptical because there have certainly been other studies trying to find this that failed. there are at least two common uses that are just beneficial. first in patients with cirrhosis and hepatorenal syndrome, spontaneous bacterial peritonitis, or large volume paracentesis, albumin repletion is helpful. Also, when using ultrafiltration to remove fluid from critically ill dialysis patients, 20 or 25% albumin is helpful in reducing hypotension. the unanswered questions on the right are still pretty relevant. Most of the studies in critical illness have used modest volumes, sometimes up to a liter of albumin but rarely much more than that. He will often see clinicians switch to albumin after a substantial volume of crystalloid, but where the patient does not quite yet meet a transfusion threshold. That is often at some arbitrary number like 2 L or 4 L. Related to that is a not well defined benefit from the Lesser quantity of tissue edema that colloid generates. the existing trials have not been in pancreatic surgery or microvascular flaps where moderate quantities of edema can be quite harmful. I think there is still valid room to debate whether or not it can play a role in those cases, especially when using 20% or 25%.

American Burn Association Clinical Practice Guidelines on Burn Shock Resuscitation

Robert Cartotto [1], Laura S Johnson [2], Alisa Savetamal [3], David Greenhalgh [4], John C Kubasiak [5], Tam N Pham [6], Julie A Rizzo [7 8], Soman Sen [9], Emilia Main [10]

Affiliations + expand

PMID: 38051821 DOI: 10.1093/jbcr/irad125

Abstract

This Clinical Practice Guideline (CPG) addresses the topic of acute fluid resuscitation during the first 48 hours following a burn injury for adults with burns ≥20% of the total body surface area (%TBSA). The listed authors formed an investigation panel and developed clinically relevant PICO (Population, Intervention, Comparator, Outcome) questions. A systematic literature search returned 5978 titles related to this topic and after 3 levels of screening, 24 studies met criteria to address the PICO questions and were critically reviewed. We recommend that clinicians consider the use of human albumin solution, especially in patients with larger burns, to lower resuscitation volumes and improve urine output. We recommend initiating resuscitation based on providing 2 mL/kg/% TBSA burn in order to reduce resuscitation fluid volumes. We recommend selective monitoring of intra-abdominal and intraocular pressure during burn shock resuscitation. We make a weak

One special population that might help inform this practice is a group of patients that receives a lot of fluid but also has some unique physiologic injuries: burn patients. you will find that many centers administer albumin either as a primary resuscitation fluid or as a rescue resuscitation fluid when patients begin to have " fluid creep". Studies in large burns have found that the amount of fluid administered and total body water are reduced with an albumin first strategy, however there is little to no reliable data that this strategy is actually beneficial for patient-oriented outcomes. on the screen is the last clinical practice guideline from 2024, which if you are interested does a very thorough job of reviewing this literature.

Studies to add

- https://pubmed.ncbi.nlm.nih.gov/30073665/ MA of 30k crit care patients in 65 RCTs: it makes no difference
- https://pubmed.ncbi.nlm.nih.gov/38101966/ - metaanalysis of 2711 patients / 15 trials in cardiac and vascular, no benefit to all cause mortality, AKI, LoS, ICULoS, blood use, CI
- https://pubmed.ncbi.nlm.nih.gov/37477023/ - no benefit in hypoalbuminemic patients
- https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9478283/ better ventilation, less hypotension postop
- https://pubmed.ncbi.nlm.nih.gov/36521076/ MPOG massive observational studies, HA *harms*
- https://pubmed.ncbi.nlm.nih.gov/37891142/ 2ndary analysis, less CK-MB with HA
- https://pubmed.ncbi.nlm.nih.gov/36749856/ small (n=134) RCT, no benefit in mostly low-MELD OLT of 20% * 200 mL
- https://pubmed.ncbi.nlm.nih.gov/39098542/ 2ndary analysis, no difference in hypotension, minimal difference in lactate at 1 time point
- https://pubmed.ncbi.nlm.nih.gov/32058387/ albumin in acute liver failure is weak recommendation (indirect data only)
- https://pubmed.ncbi.nlm.nih.gov/39846907/ early albumin in ED sepsis patients -> less vasopressor, lower SOFA (pilot, small trial)
- https://pubmed.ncbi.nlm.nih.gov/40020556/ small trial in sepsis: no improvement in patient oriented outcomes, better microcirculation measure with HA
- https://pubmed.ncbi.nlm.nih.gov/41712212/ 2025 RCT of HA in septic shock targeting 3.0, n=440 no impact

I have grown weary of adding studies of albumin.

These questions show up on your exams

In SAFE, among patients with TBI, ICP and *late* mortality was increased relative to saline. SAFE used *hypo-osmotic* 4% albumin, which probably increased brain edema. These specific products are not recommended in TBI or neurosurgery. 5% products are iso-osmotic and do not appear to increase ICP or mortality.

In older observational cohorts, albumin increased AKI. This was not replicated in randomized trials and is no longer thought to be true.

Colloid Safety

- Hetastarch: **Black Box Warning**
  - In ***ICU*** *patients*: HES increases renal replacement therapy and mortality
- Dextrans

JAMA. 2013;309(7):678-688

  - Contraindicated: active coagulopathy; renal disease; active cardiac decompensation
- Gelatins
  - Not available in the United States…

Meta-Analysis > Anesthesiology. 2026 Sep 1;145(3):591-603.
doi: 10.1097/ALN.0000000000006163. Epub 2026 May 20.

Hydroxyethyl Starch and Perioperative Complications: A Systematic Review and Meta-analysis

Daniel I Sessler [1], Lorenzo Pradelli [2], Massimiliano Povero [2], Daniel Chappell [3], E Wang [4]

Affiliations + expand

PMID: 42160629 PMCID: PMC13460315 DOI: 10.1097/ALN.0000000000006163

Results: After screening 1,114 studies, 114 trials (2009 to 2026) with 13,951 patients were included. Most trials were at low risk of bias, and there was no evidence of publication bias. Patients had major cardiac surgeries (30%), major abdominal surgeries (46%), and other surgical procedures (24%). Nearl all trials restricted HES administration to less than 24 h. HES 130/0.4 or HES 130/0.42 did not significantly increase the risk of acute kidney injury (relative risk, 1.02; 95% CI, 0.91 to 1.16) or worsen preoperative-to-postoperative creatinine concentration (-0.62 µmol/l; 95% CI, -3.82 to 2.57 µmol/l). Trial sequential analysis indicated strong noninferiority for the change in creatinine concentration and adequate information size. HES did not meaningfully worsen any other outcome evaluated, including overall or severe adverse events or mortality.

Conclusions: Current hydroxyethyl starch solutions do not provoke renal injury in surgical patients.

Starches (HES) in their current incarnation are probably safe, but cost more and have no obvious improved outcomes over crystalloid.

Effect of Intraoperative Goal-directed Balanced Crystalloid versus Colloid Administration on Major Postoperative Morbidity: A Randomized Trial

Barbara Kabon [1], Daniel I Sessler, Andrea Kurz; Crystalloid–Colloid Study Team

Collaborators, Affiliations + expand

PMID: 30882476 DOI: 10.1097/ALN.0000000000002601

Conclusions: Doppler-guided intraoperative hydroxyethyl starch administration did not significantly reduce a composite of serious complications. However, there was also no indication of renal or other toxicity.

Randomized Controlled Trial › Eur J Anaesthesiol. 2026 Jan 1;43(1):1-10.
doi: 10.1097/EJA.0000000000002307. Epub 2025 Oct 24.

Safety and efficacy of 6% hydroxyethyl starch in patients undergoing major surgery: The randomised controlled PHOENICS trial

Wolfgang Buhre [1], Óscar Díaz-Cambronero, Simon Schaefer, Martin Novacek, Marina Soro Domingo,

Conclusion: Peri-operative use of HES was noninferior to crystalloids in short-term renal function or a composite of mortality and major complications at 90 days. PHOENICS provides robust evidence that peri-operative in-label use of HES is well tolerated.

What fluid should I give?

What crystalloid?

This one is pretty easy. The overall results comparing normal saline to balance crystalloids are pretty clear: saline has a small harmful effect. Specifically, perioperatives studies in which the volume of administered crystalloid is relatively small (typically one to two liters) tend to find a negligible effect. on your exams, if you have normal saline and a balanced crystalloid, the balance crystalloid is almost always the correct answer. there will be potentially some settings in which the hypotonic nature of lactated ringers is very undesirable, such as traumatic brain injury, in which case go back to Saline

The NEW ENGLAND JOURNAL of MEDICINE

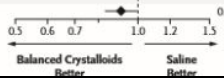
NEJM2018;378:829-39

ORIGINAL ARTICLE

Balanced Crystalloids versus Saline in Critically Ill Adults

Matthew W. Semler, M.D., Wesley H. Self, M.D., M.P.H., Jonathan P. Wanderer, M.D., Jesse M. Ehrenfeld, M.D., M.P.H., Li Wang, M.S., Daniel W. Byrne, M.S., Joanna L. Stollings, Pharm.D., Avinash B. Kumar, M.D., Christopher G. Hughes, M.D., Antonio Hernandez, M.D., Oscar D. Guillamondegui, M.D., M.P.H., Addison K. May, M.D., Liza Weavind, M.B., B.Ch., Jonathan D. Casey, M.D., Edward D. Siew, M.D., Andrew D. Shaw, M.B., Gordon R. Bernard, M.D., and Todd W. Rice, M.D., for the SMART Investigators and the Pragmatic Critical Care Research Group*

Subgroup	Balanced Crystalloids	Saline	Odds Ratio (95% CI)	P Value	P Value for Interaction
	no. of events/total no. (%)				
Unit					0.27
Medical	615/2735 (22.5)	659/2646 (24.9)	0.87 (0.77–0.99)	0.04	
Cardiac	202/1470 (13.7)	190/1501 (12.7)	1.10 (0.89–1.36)	0.38	
Neurologic	116/1440 (8.1)	141/1377 (10.2)	0.77 (0.59–0.99)	0.04	
Trauma	131/1640 (8.0)	142/1688 (8.4)	0.95 (0.74–1.21)	0.66	
Surgical	75/657 (11.4)	79/648 (12.2)	0.93 (0.66–1.30)	0.66	
Sepsis					0.06
No	744/6775 (11.0)	756/6691 (11.3)	0.96 (0.86–1.07)	0.47	
Yes	395/1167 (33.8)	455/1169 (38.9)	0.80 (0.67–0.94)	0.01	
Traumatic brain injury					0.24
No	1034/7244 (14.3)	1118/7195 (15.5)	0.89 (0.81–0.98)	0.01	
Yes	105/698 (15.0)	93/665 (14.0)	1.09 (0.81–1.47)	0.58	
Categories of kidney function					0.19
Normal	476/5596 (8.5)	514/5561 (9.2)	0.91 (0.80–1.04)	0.16	
Acute kidney injury	315/574 (54.9)	316/537 (58.8)	0.85 (0.67–1.08)	0.18	
Chronic kidney disease	301/1388 (21.7)	307/1360 (22.6)	0.95 (0.79–1.13)	0.55	
Previous renal-replacement therapy	47/384 (12.2)	74/402 (18.4)	0.61 (0.41–0.91)	0.01	
Overall	1139/7942 (14.3)	1211/7860 (15.4)	0.91 (0.83–0.99)	0.04	



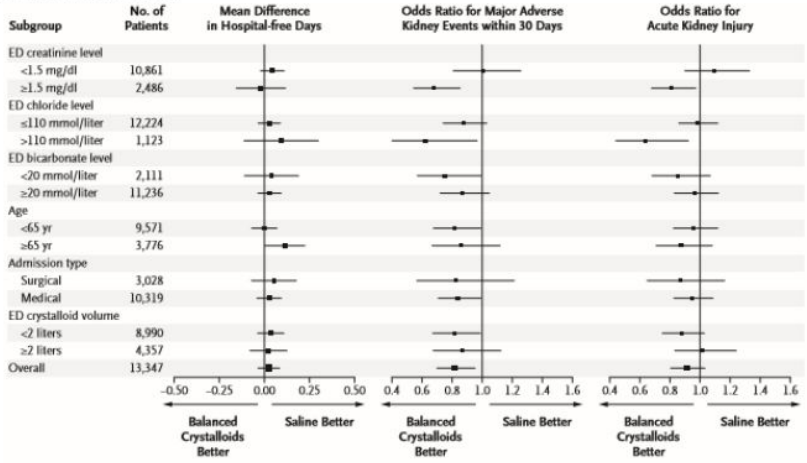
In this classical large trial there was no overall difference between the two groups, but the one subgroup in which saline appeared more harmful was in patients with prior renal failure.

ORIGINAL ARTICLE

NEJM2018;378:819-28

Balanced Crystalloids versus Saline in Noncritically Ill Adults

Wesley H. Self, M.D., M.P.H., Matthew W. Semler, M.D., Jonathan P. Wanderer, M.D., Li Wang, M.S., Daniel W. Byrne, M.S., Sean P. Collins, M.D., Corey M. Slovis, M.D., Christopher J. Lindsell, Ph.D., Jesse M. Ehrenfeld, M.D., M.P.H., Edward D. Siew, M.D., Andrew D. Shaw, M.B., Gordon R. Bernard, M.D., and Todd W. Rice, M.D., for the SALT-ED Investigators*



And the one outcome that saline appeared more harmful for was acute kidney injury

Clinical Trial > N Engl J Med. 2025 Aug 14;393(7):660-670. doi: 10.1056/NEJMoa2416761.
Epub 2025 Jun 12.

A Crossover Trial of Hospital-Wide Lactated Ringer's Solution versus Normal Saline

Lauralyn McIntyre [1 2], Dean Fergusson [1 2], Tracy McArdle [1], Shane English [1 2], Deborah J Cook [3], Alison E Fox-Robichaud [3], Claudio Martin [4], John Marshall [5], Michael Pugliese [1 6], Kusum Menon [2 7], Kednapa Thavorn [1 2], Ian D Graham [1 2], Steven Hawken [1 2], Akshai Iyengar [2], Kwadwo Kyeremanteng [1 2], Raphael Saginur [1 2], Andrew J E Seely [1 2], Ian G Stiell [1 2], Daniel Bainbridge [4], Charles Weijer [4], Monica Taljaard [1 2]; Canadian Critical Care Trials Group

Collaborators, Affiliations + expand
PMID: 40503714 DOI: 10.1056/NEJMoa2416761

Results: Seven hospitals completed both 12-week periods before the trial was interrupted owing to the coronavirus disease 2019 pandemic. Data on the primary outcome were available for 43,626 eligible patients. The mean (±SD) incidence of the composite of death or readmission to the hospital within 90 days after the index admission was 20.3±3.5% with lactated Ringer's solution and 21.4±3.3% with normal saline (adjusted difference, -0.53 percentage points; 95% confidence interval, -1.85 to 0.79; P = 0.35). Results for all secondary outcomes were consistent with those for the primary outcome. No serious adverse events were reported.

Conclusions: A hospital-wide policy to administer lactated Ringer's solution rather than normal saline did not result in a significantly lower incidence of death or readmission to the hospital within 90 days after the index admission. (Funded by the Canadian Institutes of Health Research and the

However, in patient populations who are, on the whole, receiving moderate quantities of crystalloid, it likely makes no difference, which has been observed in several large trials

Review > Br J Anaesth. 2023 Sep;131(3):463-471. doi: 10.1016/j.bja.2023.05.029.
Epub 2023 Jul 15.

Intraoperative use of balanced crystalloids versus 0.9% saline: a systematic review and meta-analysis of randomised controlled studies

Muralie Vignarajah [1], Annie Berg [1], Zahra Abdallah [1], Naman Arora [2], Arshia Javidan [3], Tyler Pitre [1], Shannon M Fernando [4], Jessica Spence [5], John Centofanti [6], Bram Rochwerg [7]

Affiliations + expand

PMID: 37455198 DOI: 10.1016/j.bja.2023.05.029

Results: Of 5959 citations, we included 38 RCTs (n=3776 patients). Pooled analysis showed that intraoperative use of balanced fluids compared with saline had an uncertain effect on postoperative mortality analysed at the longest point of follow-up (RR 1.51, 95% CI: 0.42-5.36) and postoperative need for renal replacement therapy (RR 0.95, 95% CI: 0.56-1.59), both very low certainty. Furthermore, use of balanced crystalloids probably leads to a higher postoperative serum pH (MD 0.05, 95% CI: 0.04-0.06), moderate certainty.

Intraoperative data specifically in which most patients receive a moderate amount of fluid has not been able to determine an impact on actual patient outcomes, although the effect on the pH of administering an Acidic fluid is notable.

Meta-Analysis > Br J Anaesth. 2024 Dec;133(6):1173-1182. doi: 10.1016/j.bja.2024.08.008. Epub 2024 Sep 13.

Comparative efficacy of balanced crystalloids versus 0.9% saline on delayed graft function and perioperative outcomes in kidney transplantation: a meta-analysis of randomised controlled trials

Tzu Chang [1], Ming-Chieh Shih [2], Yi-Luen Wu [3], Tsung-Ta Wu [4], Jen-Ting Yang [5], Chun-Yu Wu [6]

Results: Of 106 publications identified, we included 11 RCTs (n=1717). Pooled analysis showed that the use of balanced fluids was associated with a lower incidence of delayed graft function compared with 0.9% saline (RR 0.82, 95% CI: 0.69 to 0.98, P=0.01, moderate certainty). Balanced crystalloids were associated with higher postoperative serum pH, higher serum bicarbonate, and lower serum chloride concentration, but effects on vasopressor requirements, length of hospital stay, and serum creatinine were uncertain.

Meta-Analysis > Anesth Analg. 2024 Jul 1;139(1):58-67. doi: 10.1213/ANE.0000000000006932. Epub 2024 Apr 5.

Balanced Crystalloids Versus Normal Saline in Kidney Transplant Patients: An Updated Systematic Review, Meta-analysis, and Trial Sequential Analysis

Lucas Carvalho Pereira [1], Igor Carvalho Pereira [1], Thamiris Dias Delfino Cabral [2], Patricia Viana [3], Arthur Mendonça Ribeiro [4], Sara Amaral [5]

Results: Twelve RCTs comprising a total of 1668 patients were included; 832 (49.9%) were assigned to receive balanced solutions. Balanced crystalloids reduced the occurrence of DGF compared to normal saline, with RR of 0.82 (95% confidence interval [CI], 0.71-0.94), P = .005; I^2 = 0%. The occurrence was 25% (194 of 787) in the balanced crystalloids group and 34% (240 of 701) in the normal saline group. Moreover, our TSA supported the primary outcome result and suggests that the sample size was sufficient for our conclusion. … However, creatinine did not differ in the first (MD, -0.06 mg·dL -1 ; 95% CI, -0.38 to 0.26 mg·dL -1 ; P = .71) and seventh (MD, -0.06 mg·dL -1 ; 95% CI, -0.18 to 0.06 mg·dL -1 ; P = .30) postoperative days nor urine output in the first (MD, -1.12 L; 95% CI, -3.67 to 1.43 L; P = .39) and seventh (MD, -0.01 L; 95% CI, -0.45 to 0.42 L; P = .95) postoperative days.

A population deserves some comment is kidney transplantation. you will see many protocols that suggest alternating normal saline and balanced crystalloid. however I would say that there is more than adequate data to suggest that normal saline, because of the hyperchloremic acidosis / chloride induced renal artery constriction, tends to cause delayed graft function, although it appears that this resolves over time. The clinical significance is also going to depend on the quality of the kidney graft, with studies and trial populations that tended to have living donors finding no difference.

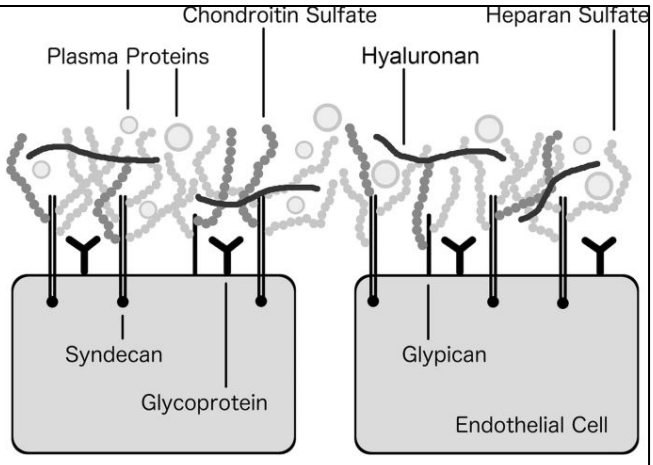
Glycocalyx

Starch / proteoglycans layer coating endothelium, regulates permeability, coagulation. Prevents reabsorption of edema outside gut.

Damaged in inflammation.

Fast crystalloids seem to worsen damage - but this may be transient in non-septic populations.

Plasma and albumin may reduce damage / regenerate.



Changes in vascular permeability dominate starling-principle type effects on fluid translocation perioperatively. Recommendations to improve care with this knowledge are very sparse.

> Anesth Analg. 2024 Aug 1;139(2):339-348. doi: 10.1213/ANE.0000000000006767. Epub 2024 Jul 15.

A Slow-Exchange Interstitial Fluid Compartment in Volunteers and Anesthetized Patients: Kinetic Analysis and Physiology

Robert G Hahn [1], Randal O Dull [2 3 4]

Affiliations + expand

PMID: 38153873 DOI: 10.1213/ANE.0000000000006767

0.080 min -1 , respectively). The return flows to the plasma were impeded by intensive fluid therapy; the rate constant for the fast-exchange compartment decreased from 0.251 to 0.050 when the infusion time increased from 15 to 60 minutes, and the rate constant for the slow-exchange compartment decreased from 0.019 to 0.005 when the infused volume increased from 500 to 1500 mL. The slow-exchange compartment became disproportionately expanded when larger fluid volumes were infused and even attained an unphysiologically large size when general anesthesia was added, suggesting that the flow of fluid was restrained and not solely determined by hydrostatic and oncotic forces. The dependence of the slow-exchange compartment on general anesthesia, crystalloid infusion rate, and infusion volume all suggest a causal physiological process.

Conclusions: Kinetic analysis supported that Ringer's solution distributes in 2 interstitial compartments with different turnover times. The slow compartment became dominant when large amounts of fluid were infused and during general anesthesia. These findings may explain why fluid accumulates in peripheral tissues during surgery and why infused fluid can remain in the body for several days after general anesthesia.

> Acta Anaesthesiol Scand. 2016 May;60(5):569-78. doi: 10.1111/aas.12686. Epub 2016 Jan 13.

Distribution of crystalloid fluid changes with the rate of infusion: a population-based study

R G Hahn [1], D Drobin [2], J Zdolsek [3]

Affiliations + expand

PMID: 26763732 DOI: 10.1111/aas.12686

Conclusion: The return of Ringer's acetate from the peripheral fluid compartment to the plasma was slower with high than with low infusion rates. Edema is a normal consequence of plasma volume expansion with this fluid, even in healthy volunteers. The results are consistent with the view that the viscoelastic properties of the interstitial matrix are responsible for the distribution and redistribution characteristics of crystalloid fluid.

In healthy volunteers there have been several studies that have validated the notion that more rapid fluid administration leads to transfer of fluid into a compartment that slowly returns to the vascular Supply.

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Effect of Intravenous Fluid Treatment With a Balanced Solution vs 0.9% Saline Solution on Mortality in Critically Ill Patients

The BaSICS Randomized Clinical Trial

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Conclusion and Relevance Among critically ill patients requiring fluid challenges, use of a balanced solution compared with 0.9% saline solution did not significantly reduce 90-day mortality. The findings do not support the use of this balanced solution.

Conclusions and Relevance Among patients in the intensive care unit requiring fluid challenges, infusing at a slower rate compared with a faster rate did not reduce 90-day mortality. These findings do not support the use of a slower infusion rate.

However in the one trial actually testing this that I know about, which was perhaps underpowered for this outcome, there was no difference in clinical endpoints between rapid and slower IV fluid administration in critically ill patients

Endothelial Glycocalyx Degradation in Sepsis: Analysis of the Crystalloid Liberal or Vasopressors Early Resuscitation in Sepsis (CLOVERS) Trial, a Multicenter, Phase 3, Randomized Trial

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DOI: 10.1513/AnnalsATS.202501-012OC

Abstract

Rationale: Prior evidence suggests that endothelial glycocalyx degradation contributes to sepsis pathogenesis and is potentially worsened by intravenous fluid resuscitation. **Objectives:** To assess *1*) the association of endothelial glycocalyx degradation with sepsis mortality, *2*) the impact of a randomly assigned liberal versus restrictive intravenous fluid resuscitation strategy on endothelial glycocalyx degradation, and *3*) whether there is a differential treatment effect for mortality based on baseline endothelial glycocalyx degradation. **Methods:** We used an enriched sampling strategy to define a cohort of 574 patients enrolled in the CLOVERS (Crystalloid Liberal or Vasopressors Early Resuscitation in Sepsis) trial, which compared liberal versus restrictive intravenous fluid resuscitation strategies. We used mass spectrometry to quantify plasma heparan sulfate as the primary measure of endothelial glycocalyx degradation. Plasma syndecan-1, quantified by enzyme-linked immunoassay, served as a clinically feasible complementary index of endothelial glycocalyx degradation. The primary outcome was 90-day all-cause mortality. **Results:** There was an association between baseline heparan sulfate level and mortality, with increasing mortality by baseline heparan sulfate tertile: lower tertile, 9.9% (95% confidence interval, 7.0-12.7%); middle tertile, 20.4% (15.6-25.0%); and upper tertile, 44.2% (35.6-51.6%) ($P < 0.001$; log-rank test), with an adjusted hazard ratio for interquartile range change in heparan sulfate of 3.12 (95% confidence interval, 2.18-4.46). We observed no effect of assigned fluid resuscitation strategy on endothelial glycocalyx degradation 24 hours after randomization. We observed no evidence of differential treatment effect of fluid resuscitation strategy based on baseline plasma heparan sulfate for 90-day mortality. Similar findings were observed using plasma syndecan-1 as an index of endothelial glycocalyx degradation. **Conclusions:** Endothelial glycocalyx degradation is a strong predictor of mortality in sepsis. Fluid resuscitation strategy had no impact on endothelial glycocalyx degradation, and there was no evidence for a differential effect of resuscitation strategy by baseline levels of endothelial glycocalyx degradation. Clinical trial registered with www.clinicaltrials.gov (NCT03434028).

Randomly assigned liberal vs restrictive fluid had *no effect* on glycocalyx degradation. Suggests that prior results on liberal fluids *causing* glycocalyx dysfunction were confounding (worse sepsis in both) or reverse causation (more edema -> more fluid demand).

Restrictive Versus Standard Intravenous Fluid Therapy and Endothelial Glycocalyx Shedding in ICU Patients With Septic Shock–A Preplanned Sub-Study of the Randomized CLASSIC Trial

Jens Christensen [1], Praleene Sivapalan [2], Tine Sylvest Meyhoff [2], Hans Järnbert-Pettersson [1], Anders Perner [2], Morten Hylander Møller [2], Theis Lange [3], Peter Buhl Hjortrup [2], Eva Joelsson-Alm [1], Sandra Jonmarker [1], Fredric Sjöberg [1], Johan Mårtensson [4], Anna Håkansson Gladh [1], Maria Cronhjort [1]

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PMID: 41321198 PMCID: PMC12666609 DOI: 10.1111/aas.70156

In this underpowered sub-study we found no statistically significant difference in endothelial glycocalyx shedding between adult ICU patients with septic shock randomized to restrictive versus standard IV fluid therapy.

Another piece of empirical damage to the glycocalyx theory from the last year was two trials of liberal versus restrictive IV fluids in sepsis, which actually included markers of glycocalox shedding. neither of them found an association between the randomly assigned IV fluid strategy and glycolic shedding, which suggests that some of the prior studies which found an association were driven either by confounding (worse sepsis in both) or reverse causation (more edema -> more fluid demand). I would say that this is still an early set of results, and keep your mind open to the possibility that this will get reversed in future analysis.

American Burn Association Clinical Practice Guidelines on Burn Shock Resuscitation

Robert Cartotto [1], Laura S Johnson [2], Alisa Savetamal [3], David Greenhalgh [4], John C Kubasiak [5], Tam N Pham [6], Julie A Rizzo [7 8], Soman Sen [9], Emilia Main [10]

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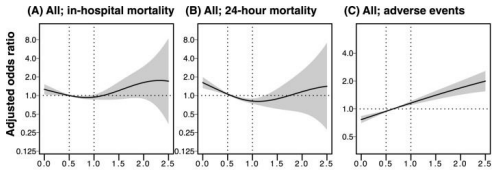
transpulmonary thermodilution-derived variables to guide burn shock resuscitation. We are unable to make any recommendations on the use of high-dose vitamin C (ascorbic acid), fresh frozen plasma (FFP), early continuous renal replacement therapy, or vasopressors as adjuncts during acute burn shock resuscitation. Mortality is an important outcome in burn shock resuscitation, but it was not

Association between the plasma-to-red blood cell ratio and survival in geriatric and non-geriatric trauma patients undergoing massive transfusion: a retrospective cohort study

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Multicenter Study > JAMA Surg. 2024 Nov 1;159(11):1272-1280.
doi: 10.1001/jamasurg.2024.3097.

High Fresh Frozen Plasma to Red Blood Cell Ratio and Survival Outcomes in Blunt Trauma

Gaku Fujiwara [1 2 3 4], Yohei Okada [5 6], Wataru Ishii [7], Tadashi Echigo [8], Naoto Shiomi [9], Shigeru Ohtsuru [4]

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PMID: 39167374 PMCID: PMC11339704 DOI: 10.1001/jamasurg.2024.3097

Conclusions and relevance: In this cohort study, a high FFP to RBC ratio was associated with favorable survival in patients with severe blunt trauma. These outcomes highlight the importance of revising the current transfusion protocols to incorporate a high FFP to RBC ratio, warranting further research on optimal patient treatment.

One other strategy that has garnered some attention in the last couple of years is the early use of plasma as potentially helping to regenerate damage to the endothelial glycocalyx. it's worth remembering that the glycocalyx is composed both of starches extruded from the endothelium and plasma proteins, and that plasma proteins play a particularly important role in protecting those starches from degradation. in animal models and some healthy volunteer studies there is fairly clear signal that plasma Administration either prevents or reverses glycocalyx damage. one important population that I mentioned earlier is that many burn centers use plasma long before it has any impact on coagulation. however, despite this practice being popular there is not adequate data to support it as improving outcomes other than fluid balance.

The other population in which there has been significant debate about the role of early plasma has been trauma. I will say that the role of high plasma ratios is not clear with different seeming results in American and Japanese data. additionally, the effect due to the impact Plasma has on the glycocalyx is difficult to disambiguate from the effect of Plasma on consumptive coagulopathy in trauma.

ABA Content outline

5. IV Fluid Therapy during Anesthesia

a. Water, Electrolyte, Glucose Requirements and Disposition

b. Crystalloid vs. Colloid

c. Fluid Requirements and Fluid Deficit Calculations

d. Normal Saline vs. Lactated Ringer's vs. Plasma-Lyte vs. D5W

e. Fluid Management and Patient Outcomes